

DIP HW3_2

313512009 施丞蔚

1. Please use a Histogram Statistics method and a local enhancement method to extract the hidden image of the image, 'hidden object.jpg.'
Please describe the your parameters and method in detail and print out the source code? (40+40)

q1.py:

```
import sys
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt
import cv2

# ----- 讀檔 -----
def resolve_input():
    if len(sys.argv) > 1:
        p = Path(sys.argv[1])
    else:
        # 常見三種命名
        cand = [
            Path("hidden_object.jpg"),
            Path("hidden object.jpg"),
            Path("hidden_object.tif"),
            Path("hidden object.tif"),
        ]
        p = next((c for c in cand if c.exists()), None)
        if p is None:
            p = Path(__file__).resolve().parent / "hidden object.jpg"
        if not p.exists():
            alt = p.with_suffix(".tif")
            if alt.exists(): p = alt
        if not p.exists():
            raise FileNotFoundError(f"找不到影像：{p}\n用法：python3
q1_show_matlab_equiv.py 'hidden_object.jpg'")
    return p

# ----- 工具：直方圖（機率） -----
def hist_prob(img_u8):
    h, _ = np.histogram(img_u8.ravel(), bins=256, range=(0, 255))
    return h / max(h.sum(), 1)
```

```

def show_pair(img_u8, title_img, title_hist):
    fig, axs = plt.subplots(1, 2, figsize=(10, 4))
    axs[0].imshow(img_u8, cmap='gray', vmin=0, vmax=255)
    axs[0].set_title(title_img); axs[0].axis('off')
    p = hist_prob(img_u8)
    axs[1].bar(np.arange(256), p, width=1.0)
    axs[1].set_xlim(0, 255)
    axs[1].set_xlabel('Gray level'); axs[1].set_ylabel('Probability')
    axs[1].set_title(title_hist)
    fig.tight_layout()
    plt.show()

# ----- 全域直方圖等化 -----
def global_hist_equalize(img_u8):
    hist, _ = np.histogram(img_u8.flatten(), 256, [0, 256])
    cdf = hist.cumsum()
    cdf_masked = np.ma.masked_equal(cdf, 0)
    cdf_scaled = (cdf_masked - cdf_masked.min()) * 255 /
(cdf_masked.max() - cdf_masked.min())
    lut = np.ma.filled(cdf_scaled, 0).astype('uint8')
    return lut[img_u8]

def main():
    in_path = resolve_input()
    # 讀 jpg/tif 為灰階
    img = cv2.imread(str(in_path), cv2.IMREAD_GRAYSCALE)
    if img is None:
        raise RuntimeError("cv2.imread 讀檔失敗, 請確認檔名/副檔名")

    H, W = 665, 652
    img = cv2.resize(img, (W, H), interpolation=cv2.INTER_AREA)

    # (1) 原圖 + 直方圖
    show_pair(img, "Original (Grayscale, 665×652)", "Histogram:
Original")

    # (2) Global Histogram Equalization (直方圖統計法)
    img_eq = global_hist_equalize(img)
    show_pair(img_eq, "Image after Histogram Equalization", "Histogram:
Equalization")

    # (3) Local Enhancement
    # 參數

```

```

k0, k1 = 0.0, 0.25
k2, k3 = 0.0, 1.0
c_gain = 3.0

# 全域  $\mu$ 、 $\sigma$ 
g = img_eq.astype(np.float32)
mG = g.mean()
sG = g.std() + 1e-12

# 3×3 區域均值與標準差 (用 box blur/blur 計算)
g32 = g.astype(np.float32)
mL = cv2.blur(g32, (3, 3))           #  $E[x]$ 
m2L = cv2.blur(g32 * g32, (3, 3))    #  $E[x^2]$ 
sL = cv2.sqrt(cv2.max(m2L - mL * mL, 0)) #  $\sqrt{\text{Var}}$ 

# 條件:  $k0 \cdot mG \leq mL \leq k1 \cdot mG$  且  $k2 \cdot sG \leq sL \leq k3 \cdot sG$ 
mask = (mL >= k0 * mG) & (mL <= k1 * mG) & (sL >= k2 * sG) & (sL <=
k3 * sG)

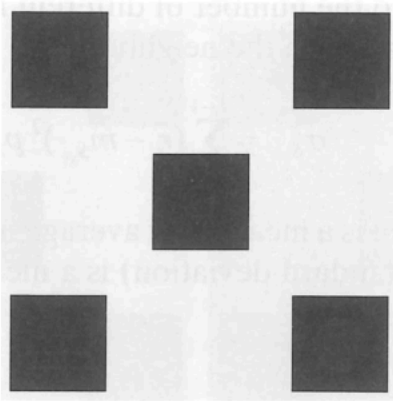
# 乘上增益 c (並截斷到 0..255)
out = img_eq.copy().astype(np.float32)
out[mask] = np.clip(c_gain * out[mask], 0, 255)
out = out.astype(np.uint8)

show_pair(out, f"Local Enhancement (3×3,
k0={k0},k1={k1},k2={k2},k3={k3}, c={c_gain})",
          "Histogram: Local Enhancement")

if __name__ == "__main__":
    main()

```

Original (Grayscale, 665×652)



Histogram: Original

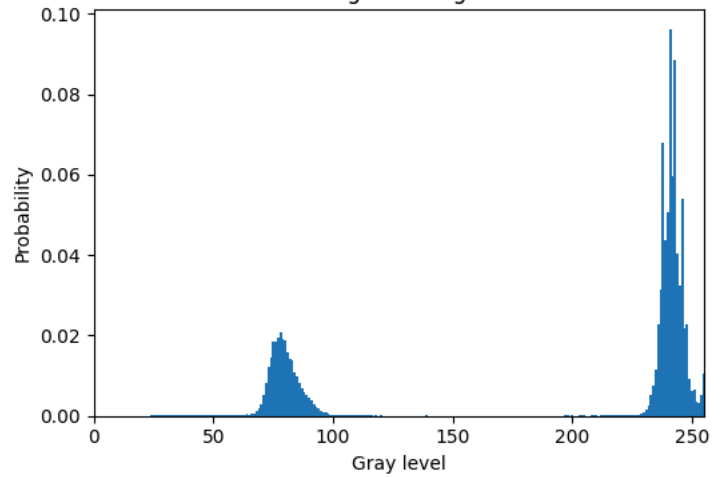
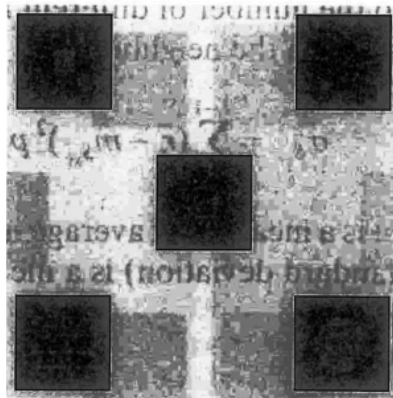
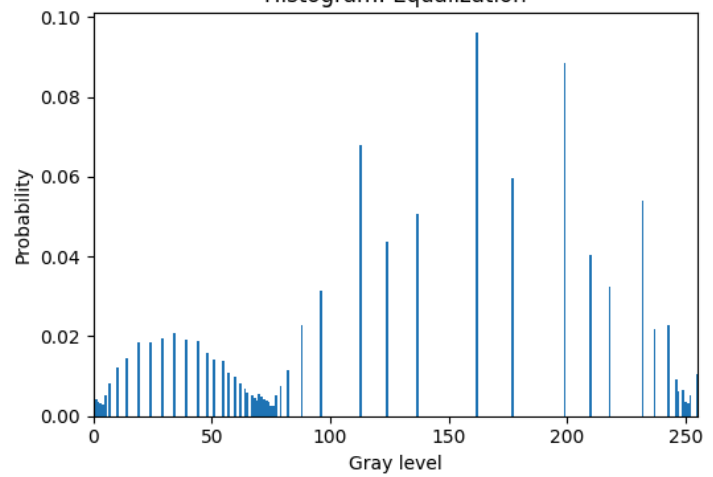


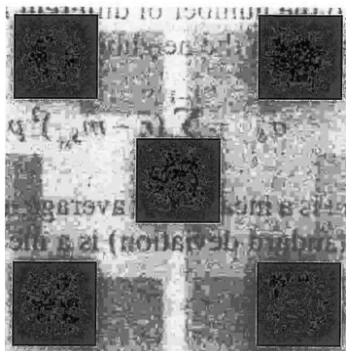
Image after Histogram Equalization



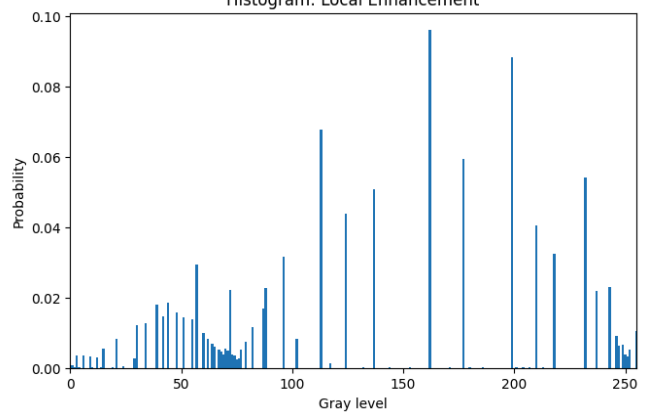
Histogram: Equalization



Local Enhancement (3×3, $k_0=0.0, k_1=0.25, k_2=0.0, k_3=1.0, c=3.0$)



Histogram: Local Enhancement



q1-2.py (另一方法嘗試):

```
# q1_show_final.py
# 直接顯示三組二合一（左影像、右直方圖）: Original / Global HE / Local
Enhancement (Dark-biased CLAHE)

import sys
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt
import cv2

# ===== 可調參數 =====
# 先做輕微中值濾波，抑制雜點 (None/0 表示不做；建議用 3)
MEDIAN_BLUR_K = 3

# 暗部遮罩的門檻與過渡帶寬（越高越多區域走「hard」）
DARK_THRESH = 150          # 120~160 常用；想更像“很兇”的參考圖可設 150~160
SOFT_BAND    = 30.0        # 過渡帶寬，越大越柔和

# CLAHE 兩組參數：soft（非暗部）與 hard（暗部）
CLAHE_SOFT_CLIP = 2.5
CLAHE_SOFT_TILE = (8, 8)

CLAHE_HARD_CLIP = 7.0      # 6~10 越大越兇
CLAHE_HARD_TILE = (4, 4)   # (3,3) 或 (4,4) tile 越小區塊感越強
# =====

def load_input_path():
    if len(sys.argv) > 1:
        p = Path(sys.argv[1])
    else:
        p = Path(__file__).resolve().parent / "hidden object.jpg"
    if not p.exists():
        alt = p.with_suffix(".tif")
        if alt.exists():
            p = alt
    if not p.exists():
        raise FileNotFoundError(f"找不到影像：{p}\n用法：python3
q1_show_final.py 'hidden object.jpg'")
    return p

def global_hist_equalize(img_u8):
    """全域直方圖等化 (Histogram Statistics / HE) """
```

```

hist, _ = np.histogram(img_u8.flatten(), 256, [0, 256])
cdf = hist.cumsum()
cdf_masked = np.ma.masked_equal(cdf, 0)
cdf_scaled = (cdf_masked - cdf_masked.min()) * 255 /
(cdf_masked.max() - cdf_masked.min())
lut = np.ma.filled(cdf_scaled, 0).astype('uint8')
return lut[img_u8]

def clahe_apply(img_u8, clip, tile):
    clahe = cv2.createCLAHE(clipLimit=clip, tileGridSize=tile)
    return clahe.apply(img_u8)

def dark_biased_clahe(img_u8,
                      soft_clip=CLAHE_SOFT_CLIP,
soft_tile=CLAHE_SOFT_TILE,
                      hard_clip=CLAHE_HARD_CLIP,
hard_tile=CLAHE_HARD_TILE,
                      dark_T=DARK_THRESH, band=SOFT_BAND,
median_k=MEDIAN_BLUR_K):
    """只對偏暗區域套用更兇的 CLAHE，其他區域用較溫和的 CLAHE，再做權重融合。"""
    base = img_u8.copy()
    if median_k and median_k >= 3:
        base = cv2.medianBlur(base, median_k)

    soft = clahe_apply(base, soft_clip, soft_tile)
    hard = clahe_apply(base, hard_clip, hard_tile)

    blur = cv2.GaussianBlur(base, (9, 9), 0)
    # 權重：越暗權重越接近 1（越用 hard）
    w = np.clip((dark_T - blur) / float(band), 0.0,
1.0).astype(np.float32)

    out = (w * hard + (1.0 - w) * soft).astype(np.uint8)
    return out

def show_pair(image_u8, title_img, title_hist):
    """顯示二合一：左影像、右直方圖（機率）"""
    fig, axs = plt.subplots(1, 2, figsize=(10, 4))
    axs[0].imshow(image_u8, cmap='gray', vmin=0, vmax=255)
    axs[0].set_title(title_img)
    axs[0].axis('off')

    h, bins = np.histogram(image_u8.ravel(), bins=256, range=(0, 255))

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p = h / max(h.sum(), 1)
axs[1].bar(bins[:-1], p, width=1.0)
axs[1].set_xlim(0, 255)
axs[1].set_xlabel('Gray level')
axs[1].set_ylabel('Probability')
axs[1].set_title(title_hist)

fig.tight_layout()
plt.show()

def main():
    in_path = load_input_path()
    print("[INFO] OpenCV:", cv2.__version__)
    print("[INFO] Input:", in_path.resolve())

    # 灰階讀檔 (jpg/tif 都可)
    img = cv2.imread(str(in_path), cv2.IMREAD_GRAYSCALE)
    if img is None:
        raise RuntimeError("cv2.imread 讀檔失敗, 請確認檔名/副檔名。")

    # (1) 原圖 (可選: 同樣做 median blur 以便公平)
    orig = img.copy()
    if MEDIAN_BLUR_K and MEDIAN_BLUR_K >= 3:
        orig = cv2.medianBlur(orig, MEDIAN_BLUR_K)

    # (2) 全域直方圖等化 (Histogram Statistics)
    he = global_hist_equalize(orig)

    # (3) 區域增強: 暗部偏重的 CLAHE 融合
    loc = dark_biased_clahe(img_u8=img)

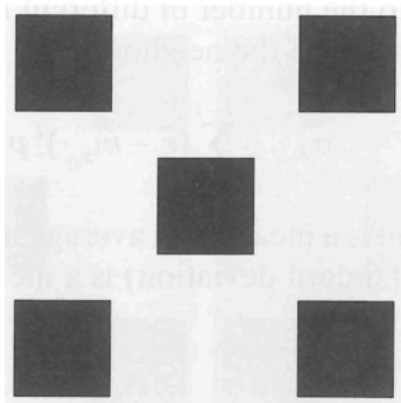
    # 依序顯示三組
    show_pair(orig, "Original (with median blur)" if MEDIAN_BLUR_K else
"Original",
              "Histogram: Original")
    show_pair(he, "Global Histogram Equalization (HE)", "Histogram:
Global HE")
    show_pair(loc,
              f"Local Enhancement (Dark-biased CLAHE)\n"
              f"soft: clip={CLAHE_SOFT_CLIP}, tile={CLAHE_SOFT_TILE}; "
              f"hard: clip={CLAHE_HARD_CLIP}, tile={CLAHE_HARD_TILE}; "
              f"T={DARK_THRESH}, band={SOFT_BAND},
median={MEDIAN_BLUR_K}",

```

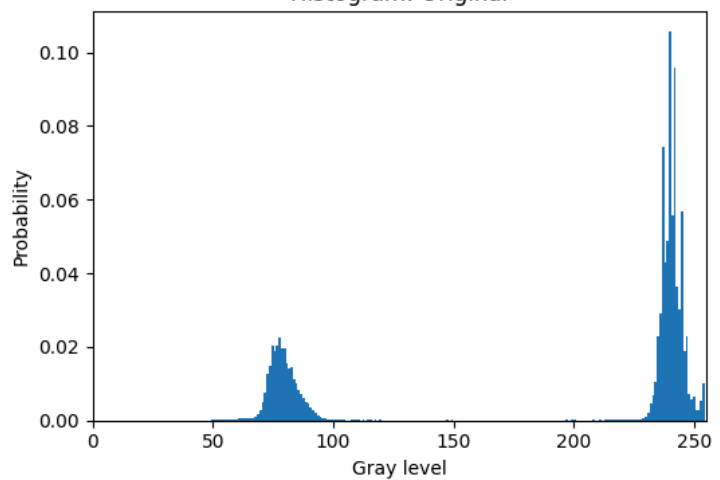
```
"Histogram: Local (CLAHE blend)")

if __name__ == "__main__":
    main()
```

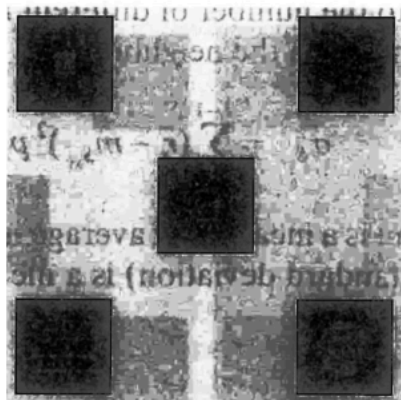
Original (with median blur)



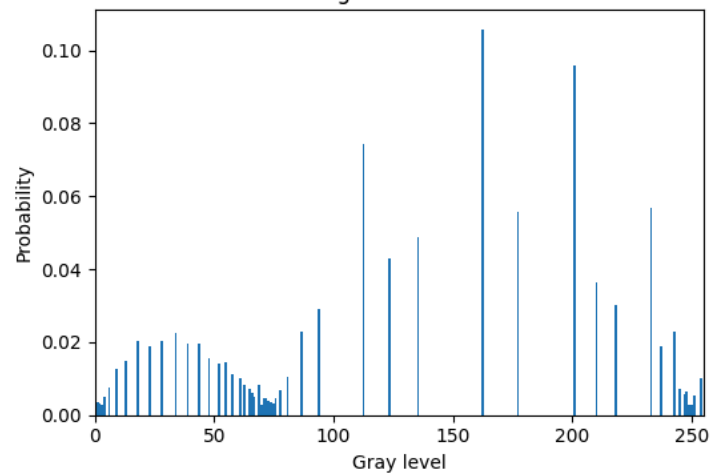
Histogram: Original



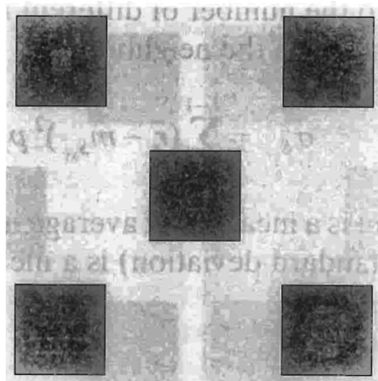
Global Histogram Equalization (HE)



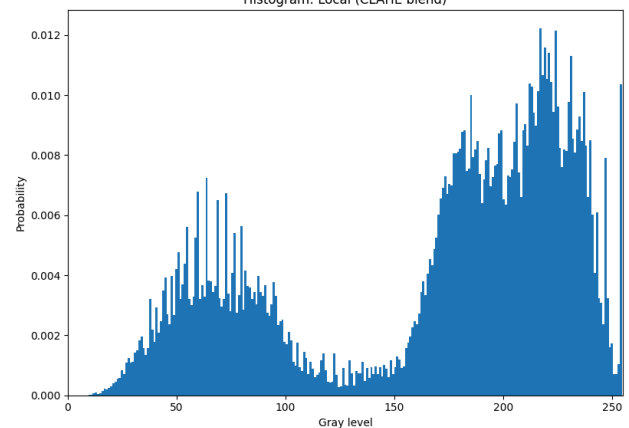
Histogram: Global HE



Local Enhancement (Dark-biased CLAHE)
soft: clip=2.5, tile=(8, 8); hard: clip=7.0, tile=(4, 4); T=150, band=30.0, median=3



Histogram: Local (CLAHE blend)



2. Please comment and compare Histogram Statistics method and a local enhancement method? (20)

q1.py:

在這張影像上，原圖的灰階直方圖明顯呈雙峰：一小峰約落在 60–90（黑色方塊與較深陰影），一大峰集中於 230–250（紙面高亮背景），中間 100–200 幾乎稀疏，因此紙背的淡字因落在狹窄的中灰區而不易辨識。

經全域直方圖等化後，所有灰階以同一映射被攤開，原本擠在高亮端的背景大量被推離 230–250，整體分布在 0–255 間出現許多離散尖峰與稀疏段；對應在畫面上，背景中灰被顯著拉出，紙背字樣幾乎全面浮現，但黑方塊與紙纖維的微小起伏也被同步放大，產生顆粒與局部飽和，方塊邊緣的粗糙感明顯增加。

區域增強法則以局部統計（ 3×3 的 $m_{\square}$ 、 $s_{\square}$ 與全域 m_G 、 s_G 的門檻）進行選擇性放大，結果在視覺上仍能把紙背文字推到可讀，同時對原本已高對比的方塊與亮區較為克制，噪點擴張較少；其直方圖主要在原本稀薄的中灰區（約 40–120 與 150–210）出現增厚與多峰，但不像全域等化那樣把整個高亮峰全面撕開，分布擴張較為「局部」。

q1-2.py:

經全域直方圖等化（HE）後，所有像素以同一 LUT 被重映射，直方圖從兩個主峰被拉展為滿布於 0–255 的多個離散尖峰；這對應到影像上，背景的中灰帶被大幅打開，紙背文字幾乎全面浮現，但同時黑方塊與紙纖維的微小起伏一併被擴張，顆粒感與局部飽和明顯，塊內噪點對比增加。

區域增強（dark-biased CLAHE 融合）則以小區塊內的等化為主，並在偏暗區域施以較大的 clipLimit／較小的 tile；其結果的直方圖呈現兩個寬而平坦的主要團塊：一個約在 45–95、另一個在 190–245，中間 150–210 區域的能量相較原圖有選擇性增厚，但不會像 HE 那樣在整個範圍形成高尖峰。影像上可見紙背文字與背景紋理被強化，但已高對比的黑方塊與高亮背景增幅相對克制，塊內紋理與粗糙度的提升低於 HE；同時，由於增強在各 tile 內進行，增強形態呈現區域性，畫面整體均衡性較 HE 好、偽影主要表現在局部等化帶來的細粒度起伏，而非全域性的顆粒放大。